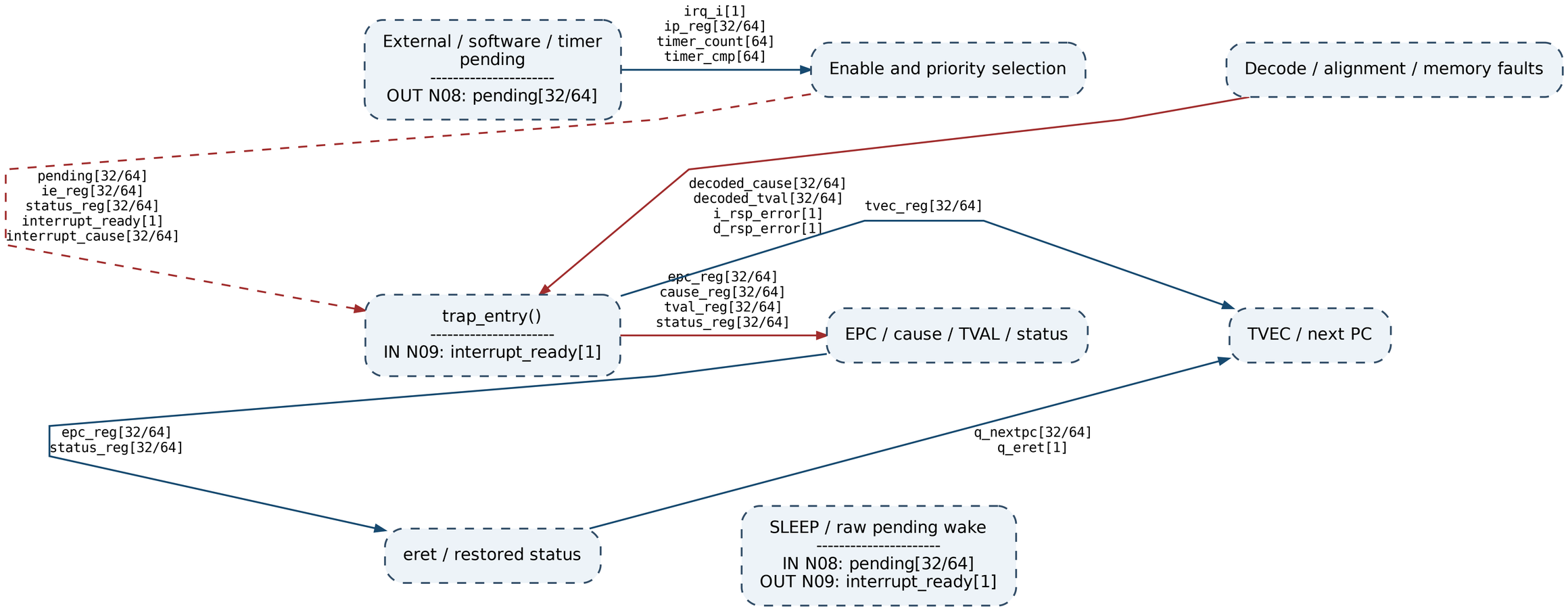
MICA 3.0 development | 09 / 20 | Precise traps, interrupt priority and wakeup

ISA 2.0 compatibility baseline | E32: XLEN=32, XREG/MUL/FLOW/BITOPS/TRAP/REG_PARITY=1 | implemented RTL

RTL source set 5fbe88e7b5ecdbe5 | internal clock: clk_i rising edge; reset: rst_ni active low

Timer pending has priority over external/software when both enabled. Global enable gates delivery, not wake.
Enabled interrupts enter directly from SLEEP; other pending events resume START.
Synchronous faults save faulting PC. 2.0 ecall also saves its own PC. No partial GPR commit.
Nested exceptions are not supported; an unrecoverable entry halts with fault_o.

Blue: data. Dashed gray: control/handshake. Red: fault. Repeated Nxx labels connect without a wire. Crossings do not join.